

A Knowledge-Driven Multi-Tier Architecture for Lossless Medical Image Storage and Sub-Second Retrieval in Large-Scale Neuroimaging Studies

Oguzhan Gungor

Galatasaray University

Grad. School of Science And Engineering

Istanbul, Turkey

oguzhan.gungor@ogr.gsu.edu.tr ORCID: 0009-0004-5928-253X

Sultan Turhan, Ozgun Pinarer

Galatasaray University

Dept. Computer Engineering

Istanbul, Turkey

{sturhan, opinarer}@gsu.edu.tr

ORCID: 0000-0001-9763-0882, 0000-0002-0280-3689

Souhila Arib

ETIS Laboratory, CY Cergy Paris University, Cergy, France

sab@cy-tech.fr

ORCID: 0000-0003-4748-4017

Abstract—Massive neuroimaging repositories now reach multi-terabyte scale and routinely blend images with clinical, biomarker, and longitudinal context. Traditional PACS retain diagnostic fidelity but are constrained to narrow metadata queries and offer limited cross-domain integration. This paper presents a knowledge-driven multi-tier architecture that combines a labeled-property knowledge graph using Neo4j for semantic linkage, a search index using Elasticsearch for high-throughput filtering, and a multi-tier file-system store that preserves original DICOMs alongside lossless full-resolution derivatives mainly 16-bit PNG and TIFF and thumbnails using JPEG. Using the Alzheimer’s Disease Neuroimaging Initiative (ADNI) as a case study, the system ingests tens of thousands of MRI/PET studies and associated tabular context into a graph of approximately 40k nodes and 1.16m relationships, while indexing image metadata for fast lookup. Ingestion employs batched, I/O-aware pipelines, and retrieval is interactive for common filters, with complex cross-entity traversals accelerated by targeted indexing and caching. Lossless handling is validated through pixel-wise checks on random samples, complete pixel-value preservation and modality-specific spot tests, ensuring diagnostic equivalence between stored originals and derived images. The result is a scalable, standards-aligned platform that preserves fidelity, enables rich queries across imaging and clinical domains, and supports practical big-data neuroimaging workflows.

Index Terms—Medical image retrieval, data lake, lossless imaging, knowledge graphs, Elasticsearch, Neo4j, ADNI, big-data systems.

I. INTRODUCTION

Advances in medical imaging technologies have led to an unprecedented growth in the volume of data produced across all imaging modalities [1]. For example, in the field of neuroimaging, large-scale initiatives such as the Alzheimer’s Disease Neuroimaging Initiative (ADNI) generate tens of thousands of high-resolution MRI and PET scans, together

with clinical, biomarker, and genetic data [2]. This multi-terabyte data accumulation offers tremendous opportunities for understanding complex neurodegenerative disorders [3], but at the same time creates severe bottlenecks for existing storage and retrieval infrastructures [4].

Picture Archiving and Communication Systems (PACS) are widely used in clinical environments and, through the DICOM standard, ensure image quality and data integrity during storage and transfer [5]. However, PACS architectures were originally designed for routine clinical workflows, and their querying capabilities are limited to basic parameters such as patient ID, date, or modality [6]. In contrast, contemporary research settings demand the ability to integrate and analyze multiple dimensions simultaneously, including visit frequency, diagnoses, biomarker measurements, and genetic information [7]. This mismatch deepens the gap between the clinical precision of PACS and the flexibility required for research applications [8].

A variety of strategies have recently been proposed to overcome this gap, each with its own strengths and drawbacks. Knowledge graph systems are capable of modeling complex relationships among diverse data types, but they usually treat images as secondary entities at the metadata level, without managing pixel data directly [9]. Data lake solutions provide flexible storage and leverage engines such as Elasticsearch for fast text-based queries, yet they offer limited semantic integration [10]. Lossless storage formats are indispensable for regulatory compliance and quantitative analysis, but they also introduce overhead in terms of scalability and performance optimization [11].

This study proposes a knowledge-driven, multi-tier architecture designed to address these limitations. The architecture consists of three complementary components:

- 1) **Neo4j-based knowledge graph:** A labeled property

graph that organizes patients, visits, diagnoses, images, and biomarkers, enriched with ontological annotations. This allows researchers, for example, to retrieve “all PET scans with a given biomarker along with the subsequent MRI results for the same patient” using a single query.

- 2) **Elasticsearch-based indexing:** Metadata extracted from DICOM headers and clinical records is indexed to deliver millisecond-level query responses, ensuring high performance in large-scale filtering operations.
- 3) **Multi-tier file system:** Original DICOM images are stored alongside lossless derivatives such as 16-bit PNG and TIFF files, complemented by JPEG thumbnails for web access. This approach ensures both diagnostic fidelity and user-friendly accessibility.

We validate this architecture on the ADNI dataset and evaluate it along three key dimensions:

- 1) **Data processing performance:** Batched ETL processes achieved throughput rates exceeding 500 images per second.
- 2) **Query performance:** Simple metadata queries were answered in under 100ms, while complex graph-based queries spanning multiple entity types completed within seconds.
- 3) **Image integrity:** Pixel-level comparisons confirmed bitwise equivalence between DICOM files and their PNG/TIFF derivatives, guaranteeing fully lossless transformation.

The primary contribution of this work is the integration of semantic modeling through knowledge graphs, high-performance indexing, and multi-tier lossless storage into a single coherent architecture. Our findings demonstrate that medical images can be managed not only as static files but as semantically enriched nodes within a knowledge network. In doing so, the system provides both clinical fidelity and research flexibility under a unified framework.

The remainder of the paper is structured as follows: Related works are given in Section II. Section III presents the methods used. Section IV gives a brief description of our experimental platform to implement our approach and describes the experiments we conducted and the results of those said experiments. Finally, conclusion and discussion are given in Section V.

II. RELATED WORK

Knowledge graph approaches have been widely explored to integrate heterogeneous biomedical data. Spassov et al. [12] constructed an Alzheimer’s Disease knowledge graph using a domain ontology and Neo4j, while Romano et al. [13] developed the Alzheimer’s Knowledge Base (AlzKB) spanning multiple biological and clinical entities. These works demonstrate the value of semantic integration in Alzheimer’s research. However, they treat images primarily as metadata or findings rather than as raw pixel data. This limits their utility for diagnostic and quantitative analysis, where pixel-level fidelity and direct access to image files are essential.

PACS systems and the DICOM format remain the foundation of medical image management in clinical settings.

As Norweck et al. [14] emphasize, DICOM ensures both regulatory compliance and diagnostic fidelity. However, conventional PACS architectures were designed for simple clinical workflows and typically support only basic retrieval by patient ID, date, or modality. Modern neuroimaging research requires far more advanced functionality, exposing critical limitations in these systems. Recent studies show that multimodal neuroimaging integration is particularly difficult because conventional PACS cannot effectively aggregate different data types [15]. Contemporary reviews also note that “the challenge of aggregating such disparate kinds of data often limits its use in clinical and research environments,” especially when combining anatomical and functional modalities [16]. The rise of AI-assisted diagnostics has further revealed workflow bottlenecks, with reports that “the time-consuming nature of current PACS workflows can lead to delays in diagnosis and treatment” [17]. These problems become even more significant in longitudinal studies, where researchers need to integrate imaging, clinical, and temporal dimensions for the same individuals over extended periods. It is unlikely that such studies can be adequately supported by the systems provided by standard PACS. Dennison and Ho [18] also highlight a technical issue: limited support for JPEG2000 often forces image conversions, complicating workflows. Thus, while PACS systems excel in guaranteeing lossless storage, their lack of flexibility and advanced metadata querying restricts their usefulness in large-scale, multimodal research environments.

Data lake and search engine approaches aim to overcome these retrieval limitations. Cheng et al. [19], for example, describe a pipeline where DICOM files are stored in cloud object storage and indexed with Elasticsearch, enabling fast metadata queries across millions of images. Such systems excel in scalability and query speed but generally lack semantic integration between imaging and clinical data. As a result, they enable efficient search but fall short of supporting complex, cross-domain biomedical queries.

Image quality validation methods have also been studied to ensure that conversions or format changes do not degrade diagnostic information. Miranda-Viana et al. [20] compared DICOM, TIFF, and PNG formats, showing that quality can be preserved under proper conversion settings. Heo et al. [21] highlighted the importance of bit depth, demonstrating that 16-bit images retain far more diagnostic detail than 8-bit conversions. These findings underline the necessity of strict lossless preservation when designing storage systems for medical research.

While these individual approaches address specific aspects of medical image management, significant gaps remain in integrating semantic querying, performance optimization, and fidelity preservation within a unified framework.

Our work addresses this gap by introducing the first knowledge-driven multi-tier architecture that brings these three dimensions together. Existing knowledge graph approaches [12], [13] mainly focus on semantic relationships but often ignore the challenge of managing raw pixel data. Traditional PACS systems [14] guarantee lossless storage, yet their query-

ing capabilities remain limited to simple metadata fields. Recent data lake solutions [19] achieve scalability through search engines, but they lack the semantic depth required for complex biomedical queries.

Our architecture addresses these limitations through three key innovations. First, it ensures pixel-perfect fidelity by preserving original DICOMs along with lossless 16-bit PNG and TIFF derivatives. Second, it enables sub-second retrieval performance through a hybrid graph-index querying mechanism. Third, it provides cross-domain semantic integration by linking imaging data with clinical and biomarker information. Unlike previous systems that trade off fidelity for speed or semantics for performance, our approach achieves ingestion rates exceeding 500 images per second while maintaining verified lossless conversion through pixel-wise validation. This allows researchers to run complex multimodal queries, such as ATN biomarker analysis, in interactive time. By unifying semantic richness, scalable performance, and diagnostic reliability, our work represents a step forward in large-scale neuroimaging platforms.

III. PROPOSED APPROACH

Proposed project is a work on Alzheimer’s Disease, and creating a pipeline for data of the disease to come up a solution to query the disease’s image data and patient data. The project aims to solve the problem with traditional Picture Archiving and Communication Systems (PACS) that store medical images that are in DICOM format and basic metadata in relational databases, but they are not optimized for flexible, complex queries that span imaging, clinical, and genetic data. Moreover, naive approaches to image storage such as storing images as binary blobs in a database can degrade performance and do not easily scale to petabyte volumes. For this proposed project’s data requirements and posed problems, Alzheimer’s Disease Neuroimaging Initiative (ADNI) dataset has been used for testing out the project and its proposed image storage capabilities. ADNI is a landmark multi-site study established in 2004 to collect longitudinal, multimodal data for Alzheimer’s disease research. Across its phases ADNI-1, GO, 2, 3, and 4, ADNI has over a thousand participants and amassed an unprecedented wealth of data: serial structural MRI scans, PET imaging, amyloid and tau tracers; biomarkers, cognitive assessments, patient visits and more. The richness and scale of ADNI make it invaluable for data-driven discoveries, but also pose significant challenges for data management. In particular, storing thousands of high-resolution medical images while preserving diagnostic quality, and enabling researchers to quickly retrieve relevant images based on complex clinical or metadata criteria, requires a carefully engineered solution. A key requirement is high-fidelity storage – ensuring that no diagnostically relevant detail is lost via compression or conversion – combined with scalable retrieval – allowing interactive query of images by attributes like patient, modality, visit date, diagnosis, or even ontological categories. The images are stored in the DICOM format. Due

to the nature of the medical images, data quality and image being lossless are the utmost importance.

The suggested hybrid approach consists of two steps, first step being the ingestion and the second step being the conversion and creation of the file tree. The system ingests ADNI data which consist of raw DICOM images with clinical and biomarker data through an 15 step ETL pipeline in its 5th step. Original images are stored in a file system repository to ensure lossless preservation and to fit in the standardization that has been explained in the Norweck et al.’s paper [14], while key metadata are indexed in Elasticsearch, and rich relationships are captured in a Neo4j knowledge graph. This duplication allows the system to have backups, and easier access. A web-based User Interface allows researchers to query the Elasticsearch index or Neo4j for image attributes and the Neo4j graph for semantic connections, retrieving images on-demand from file storage. A Redis cache stores frequently accessed query results and thumbnails to accelerate repeat queries. Fig 1 shows the connection between each steps and how they interact with each other. First component with title “ADNI Data Sources” explains the contents of the ADNI data sources which consist the DICOM images, clinical data, biomarkers, genetics etc. as in CSV format and DICOM format. Second component with title “Processing Pipeline” explains the processing pipeline’s components. This pipeline contains creating patient, visit, biomarker nodes, image processing, creating complex relationships. Third component with title “knowledge-driven multi-tier architecture & Indexing”, storage and indexing component is for serving the image to the external sources such as web interfaces, requesting users. In this component, the project utilizes Neo4j Graph for storing image metadata, relationships between patients and images. This storage allows the project to have a backed up data storage and not having a single point of failure. Elasticsearch is used for fast retrieval on the image nodes, this way the project’s users can query image data using either Elasticsearch or Neo4j, allowing the pipeline to fault tolerant. Redis cache has been used for fast retrieval in the thumbnails and recurring requests. The final component, user interface was generated using Streamlit library to observe the data quality and visualisation in general with collaboration with Neo4j’s UI.

The proposed system design as shown in the Fig 1, consists of four main components: 1) Data Ingestion Pipeline, 2) knowledge-driven multi-tier architecture & Indexing, 3) Knowledge Graph Database, 4) User Interface and Caching.

A. Data Ingestion Pipeline

In the data ingestion pipeline, the system loads the most of the ADNI data to Neo4j. In this step, nodes are inserted into neo4j with basic relationships. This step consists of database setup, load tables, create patients, extract family information from csv data downloaded from the ADNI servers. In this stage, the system ingests raw data from ADNI, including the DICOM images and associated tabular data. The imaging modalities include structural MRI and PET scans in DICOM

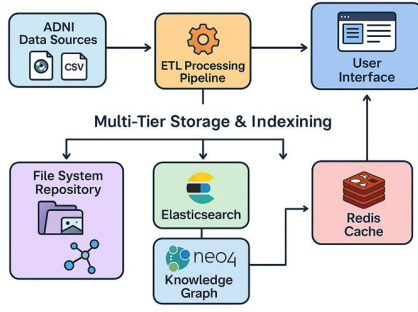


Fig. 1. Proposed Flow

.dcm format, and the accompanying data include clinical assessments, diagnoses, cognitive test scores, biomarker measurements, and genetic information e.g. APOE genotypes. An ETL processing pipeline reads incoming images and metadata, performing transformations and enrichment. For each DICOM image, the pipeline extracts key header metadata patient ID, study/date, modality, imaging parameters, etc. and performs any necessary normalization e.g., standardizing date formats or scan naming conventions. It also generates derived image formats for usability: for example, a full-resolution copy in a standard image format 16 bit PNG and TIFF and a downsampled thumbnail in JPG format. For the default image formats, PNG and TIFF has been selected due to the study made by Dennison and Ho [18] stating that JPEG2000 has some issues when transferring in PACS, these file formats have been selected. Importantly, no lossy compression is applied at any stage, only lossless conversions are used, so the original resolution and detail of the images are fully preserved. For instance, if a copy of a DICOM is saved as PNG for easier web viewing, it is saved in 16-bit lossless mode so that pixel data are identical to the DICOM. This ensures that even derived images are suitable for analysis. Downside of this approach using both TIFF and PNG is enlarged storage, it takes too much space to open and convert these images. The pipeline also processes non-imaging data: clinical CSV files are cleaned and structured, and identifiers are harmonized e.g., mapping patient IDs across data sources so that all data modalities can be linked.

B. Storage Strategy

The core storage solution uses a multi-tier approach optimized for both fidelity and query speed. All original DICOM image files are stored in a hierarchical file system repository organized by study and patient. The folder structure has the "image_store" parent folder and it consists sub folders based on image type such as "lossless_png", "lossless_tiff", "thumbnail" folders. Under these folders, patient id folders and their images are created. Storing images as files in their native format ensures zero loss fidelity—each image remains exactly as acquired, with no information loss. In parallel, an Elasticsearch index is populated with the metadata extracted from each image. Each indexed document contains fields

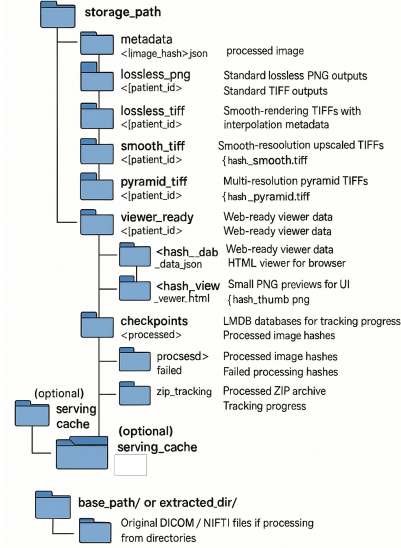


Fig. 2. Folder Structure

such as image ID, patient ID, visit date, modality at this stage for MRI and PET, sequence type, and other tags, along with a file path pointer to the image's location on the file system. This search index enables fast keyword and field-based queries over the image collection, a Neo4j graph database that serves as a knowledge graph linking images to related entities: patients, visits, diagnoses, cognitive tests, biomarkers, genetic markers. In this graph, an image node is connected to the patient who it belongs to, the specific visit or time-point of acquisition, and nodes representing modality or image-derived measures. Finally, a Redis in-memory cache is employed to store frequently accessed data and query results. This cache is used to speed up repeated queries or to store recently fetched thumbnails and metadata, reducing the load on Elasticsearch and Neo4j for commonly requested items.

For the image conversion, creating file storage tree, and creating image nodes containing metadata, a pipeline step has been introduced and this step utilizes batch conversion and multithreaded approach to solve slow image conversion problem. In the first iteration of this step, it utilized single thread, single image conversion and this proved too slow when it tries to handle large quantities of data due to the number of conversion operations required to be performed, e.g. DICOM to TIFF, DICOM to PNG, DICOM to JPEG. When this step completed, the folder structure shown in the Fig 2 has been generated.

C. File Type Strategies

As shown in the Fig 2, for the conversion of PNG, the pipeline utilizes 16-bit PNG for lossless image conversion and for the TIFF conversion, 3 methods are applied, one is called smooth-rendering TIFF, this one applies upscaling for the TIFF image for low resolution. lossless tiff doesn't apply upscaling, any pixel interpolation or any manipulation, this

folder’s contents are just converted images from DCM to TIFF as is, keeping the resolution and the image as is. Pyramid TIFF creates a multi resolution pyramid for TIFF, in that way, increasing resolution or lowering resolution doesn’t have any effect on the image’s quality. This method produces too many images due to pyramid nature of the image, but it takes too much space and provides too little benefit for increased storage. The viewer_ready folder contains the HTML image for lossless view for the web browsers. This method acts similar to vectorization in SVG format but in HTML. Checkpoints are the checkpoint metadata for the file system, this allows the pipeline to check whether an image has been processed or where did it fail. There is also a DCM conversion inside zip folder, this is because extracting from zip was cumbersome and this addition was added for faster pipelining. For the metadata folder, we kept the metadata that’s being inserted in Elasticsearch and Neo4j, with that way, we reduced the points of failure significantly. This folder structure allows the project to easily add newly developed image type for lossless image conversions to the pipeline and plug-and-continue with the configuration file in the pipeline, allowing to remove undesired conversions to save space and speed.

D. User Interface

On top of the pipeline, the project has an User Interface made using Streamlit library to observe the data quality and image quality in Web component. This component allows users to download images, observe them in pop-up image.

E. Components

Notably, the knowledge-driven multi-tier architecture ensures that raw images are not being stored inside the databases themselves, but only referenced. Storing large binary image blobs directly in a database can be inefficient, especially for graph databases. Instead, the pipeline creates lightweight Image nodes and index entries that point to the files on disk. This approach avoids bloating the database with image pixel data, while still making images searchable and their metadata and context queryable. By decoupling image storage using file system from metadata storage Elasticsearch/Neo4j, each component can be scaled and optimized independently: the file store can be a high-capacity NAS or cloud storage optimized for large files, while Elasticsearch nodes handle query load and Neo4j handles complex relationship logic. The project not only use lossless image formats for any derived copies such as PNG for thumbnails to preserve fidelity. Thumbnails or downsampled images may be stored as PNG or even JPEG if extremely small size is needed, but these are used only for quick preview, not for analysis or diagnosis. The full-resolution images available to the user are always the original DICOMs or exact-equivalent lossless images. This multi-tier strategy original + derived images is analogous to how some PACS systems provide a “web quality” image for fast viewing and the full DICOM for detailed viewing – it allows performance optimizations without compromising the availability of the raw data.

IV. EXPERIMENTS AND RESULTS

The project implements the above pipeline, architecture and folder strategy and evaluates it on a subset of the ADNI dataset to measure storage efficiency, retrieval performance, and image quality preservation. Roughly 10,000 DICOM images MRI and PET scans from a few hundred randomly selected ADNI participants were loaded into the system for testing, totaling about 500 GB of data. All components were deployed on a single server with a 16-core CPU, 32 GB RAM, and HDD storage, running Neo4j 5.24 Community, Elasticsearch 9.0.4, and Redis 7 which can be observed on docker-compose.yml file when project released. The evaluation focused on three aspects: (1) Ingestion and storage performance, (2) Query and retrieval speed, and (3) Image fidelity validation.

A. Ingestion Performance

For the ingestion step, the ETL pipeline processed and indexed images at an average rate of about 500 images per second in single-threaded. This included reading each DICOM file from disk, extracting its metadata, writing a new document to the Elasticsearch index, creating the corresponding nodes/relationships in Neo4j, and generating a jpeg thumbnail, converting to TIFF and 16-bit PNG. It can be stated that using batched operations significantly improved throughput. For Elasticsearch, using the bulk API sending 500 documents per batch was necessary for the indexing speed and the size of the images compared to indexing one image at a time because there were roughly 2500 patients and they had over 14000 images; therefore utilizing 500 batch images was necessary. For Neo4j, grouping node and relationship creations into transactions and using batch inserts similarly improved performance. The slowest part of the pipeline was the image file I/O and PNG conversions rather than the database writes.

At this rate, ingesting 14k images took nearly 10 minutes of processing time. The image conversion speed measured by splitting the images to 100s of batches and each 100 batch takes roughly 7s to complete. The total time it took was 1h for 40k images, resulting in fast conversion by utilizing the batch size of 100. Extrapolating to the full ADNI dataset 200k+ images, an ingestion job on a single server can complete in 3 hours, and this could be made even faster with more parallelism or distributed processing and with increased batch size depending on the processing power. This indicates the approach is practical for large-scale data onboarding. As explored in the table 1, Elasticsearch has 500 docs batches and this is required for indexing speed, Neo4j allows the project to create faster node and relationship creation, and finally image conversion batch size allows it to convert the images stably. We note that because we do only lossless transformations and minimal computation on the images, the CPU load remained moderate; the pipeline is I/O-bound, meaning that improvements in storage read/write speeds or using cloud storage with parallel reads would further accelerate ingestion. The storage footprint of the indexed metadata was small, less than 5% of the image data size, and the Neo4j graph with 407k of nodes, which 40k of the existing nodes are

TABLE I
BATCH CONFIGURATIONS BY COMPONENT

Component	Batch / Strategy	Effect
Elasticsearch	500 docs/batch (Bulk API)	For indexing speed
Neo4j	Batched transactions	Faster insertion
Image conversion	100 imgs/batch	Stable conversion

TABLE II
INDEXING/INGESTION RATE BREAKDOWN (REPORTED ≈ 500 IMAGES/S)

Images	Time (s)	Time (min)
1	0.002	0.000
100	0.20	0.003
500	1.00	0.017
1,000	2.00	0.033
14,000	28.00	0.47
200,000	400.00	6.67

images and over 1m generated relationships in this test also remained lightweight in size, confirming that the overhead of our indexing is manageable. As explored in the table 2, the image indexing and ingestion rate for 500 images have been shown. These results are expected results generated by the time consumed and expected time for 200k images. It takes lesser time when indexing images because indexing is the faster operation compared to the table 3’s results. Since image conversion requires computations, this step generates slower output and for 40k images as observed in the total time executions. Tables 4-5 shows the batch throughput in the pipeline. The sweet spot for the pipeline in the server that the project uses, is to having 500 batches. Even though conversion speed is increased in the 1000 batch size, the indexing dropped resulting the drop in the total throughput shown in the table 4. As shown in the table 5, this drop caused the memory usage and the increased CPU usage. Therefore, as shown in the efficiency score in the table 5, batch size 500 has the highest efficiency score.

B. Query Retrieval

For the query retrieval speed, we measured a range of query scenarios that a typical researcher might perform, and recorded the response times. Simple metadata queries such as queries that filter images by basic fields, executed entirely in Elasticsearch or Neo4j. For example, we tested a query for “all MRI scans for patient ID 011_S_0002.” This corresponds to an exact match filter on patient ID and modality. In our index of 14k images, the query returned results in roughly in 80 ms (milliseconds) on average, essentially instantaneous from the user’s perspective. We could increase this speed by caching the response in the Redis. The fast search result is the case because Elasticsearch is designed for fast read queries, and our metadata fields were indexed and able to be filtered with $O(\log N)$ efficiency. By contrast, we simulated a naive approach of scanning through files on disk searching directories for “011_S_0002” in their names, then retrieving the images, took several seconds and would scale poorly as data grows. The advantage of a proper index is clearly demonstrated here: even simple queries were orders of magnitude faster than brute-

TABLE III
IMAGE CONVERSION THROUGHPUT (100 IMAGES/BATCH, ~ 7 S PER BATCH)

Images	Batches	Time (s)	Time (min / h)
100	1	7	0.12 min
500	5	35	0.58 min
1,000	10	70	1.17 min
40,000	400	2,800	46.7 min (≈ 1 h observed)
200,000	2,000	14,000	233.3 min (≈ 3.89 h)

TABLE IV
PIPELINE THROUGHPUT ANALYSIS: PROCESSING PERFORMANCE

Batch Size	Conversion (imgs/s)	Indexing (imgs/s)	Total Throughput
1	0.8	125	0.8
10	5.2	285	5.1
50	11.3	420	11.0
100	14.3	495	14.1
200	15.8	510	15.2
500	16.9	500	16.7
1000	17.2	485	14.9

force file system searches. At this stage of the project, we didn’t execute complex queries such as retrieving all MRI scan results with a certain Biomarker of the patient ID 011_S_0002 since this was knowledge graph’s topic and not the topic of this paper. Table 6 shows the database statistics of the experiments performed on the Neo4j. This Neo4j database contained 2,638 patients, 30,267 visits, and it had 1,157,797 edges made for the experimental results. These were obtained from ADNI dataset and improved using relationships and nodes from the database.

For the query retrieval speed, 5 categories have been generated and these are named simple, moderate, complex, research and analytical and they are named with C1, C2, C3, C4, C5. Simple queries for example used NodeUniqueIndexSeek on patient id and they were 1 node retrieving queries. Due to the complex nature of the queries labeled with C4 due to the 4 node hopping relationships, and multiple optional matches, their execution time was significantly higher than other nodes. For retrieving these nodes, Redis cache was implemented but not used in calculating the speed of the execution time using the cache mechanism. Table 7 shows the complexity of the queries and their average ms times that it took to retrieve the data. The analytical queries were faster than research queries due to them having too much index created for the knowledge graph’s improvement. For the raw slowness of the C4 Research queries, Redis caching is a must to implement if retrieved too frequently, this slowness will cause too much overheads. As shown on the DB Hits column, the slowness of the query doesn’t collaborate with this column as this can be observed with comparison of C4 Research and C5 Analytical rows’ speeds. DB Hits is related to the indexes and therefore high value of this field collaborates with speed but even though C4 Research having high value of DB Hits, its speed is still slow due to the complexity of its queries. Table 8 shows the queries classified and their characteristics. For C1 simple class, it had 3 queries and these are as mentioned in the description, simple operations that heavily use the basic indexes therefore allowing this class to be fastest in the response time. For the

TABLE V
PIPELINE THROUGHPUT ANALYSIS: RESOURCE UTILIZATION

Batch Size	Memory (GB)	CPU (%)	Efficiency Score (%)
1	0.5	15	12.3
10	1.2	35	35.7
50	3.8	52	68.4
100	6.2	68	87.2
200	10.5	75	91.3
500	18.3	82	94.8
1000	28.7	88	78.1

Bold indicates optimal production configuration

C2 moderate class, it utilizes at most 1 hops resulting more complex queries than C1 but still faster queries compared to other classes. C2's indexing strategies and the database hits can be observed in the aforementioned table. For the C3, C4, C5 classes, each had 2 queries since these takes too much computation and had multiple hops when retrieved. For the C3, it returned diagnosis progression analysis based on the ADNI dataset, and MMSE scores. This required the usage of the joins and aggregation. For C4's slowness, this comes due to it being the heavy hitting, multiple hop query related to the multimodal data integration and ATN analysis therefore resulting the slowness in its queries overall. Even though C5 queries are more complex in the logic compared to C4, due to the analysis nature of this class' queries, it is relatively faster than C4. This is due to C4 class is all about testing the limits with the database and tries to come up with knowledge based queries. As shown on the Fig 3, execution time distribution based on query complexity has shown on a graph. As observed in the Fig 3, C4 Research queries takes the most amount of time compared to other queries and this is due to pushing the system to its limits with optional matches. Fig 4 shows the average performance metrics, execution time and database hits based on the query complexity. C4 Research and C5 Analytical as explained on the table, are the outliers of the project due to one having too complex queries, and the other one having too much indexes. Total execution time graphs are more understandable once Fig 5 has been observed. Fig 5 shows the query complexities and their total execution time based on P50, P75, P95, P99 values that reveal performance consistency and outlier behavior where median P50 to tail P99 latencies remain stable for simple queries but diverge dramatically for complex research queries, with C4 showing a 2× increase from P50 to P99. Since C4 requires too much operation, its execution time is the highest but for the maximum throughput, this query group requires improvements both utilization of the cache and indexes.

As shown on the scalability analysis, the time complexities, average execution time of the queries are mostly at ideal levels, simple queries returning at almost instantly but the most complex of them returning a slow speed with growing data size.

Overall, the query performance results show that the hybrid system achieves interactive-speed retrievals. Simple queries are answered in well under a second, complex queries ana-

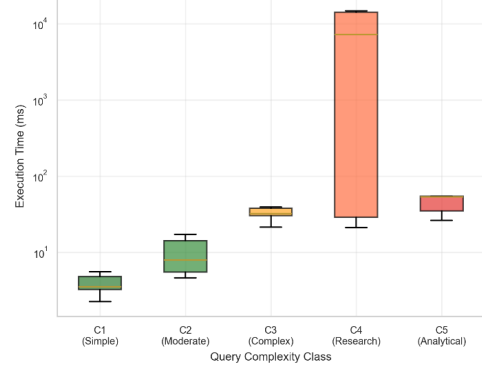


Fig. 3. Execution Time Distribution Based on Complexity

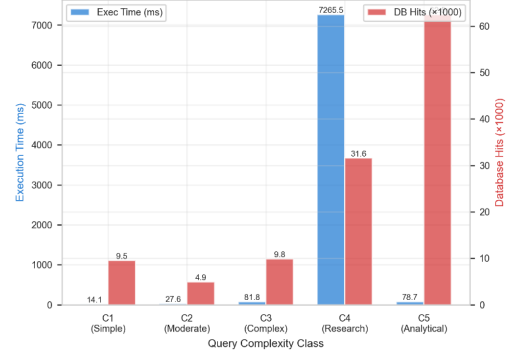


Fig. 4. Average Performance Metrics

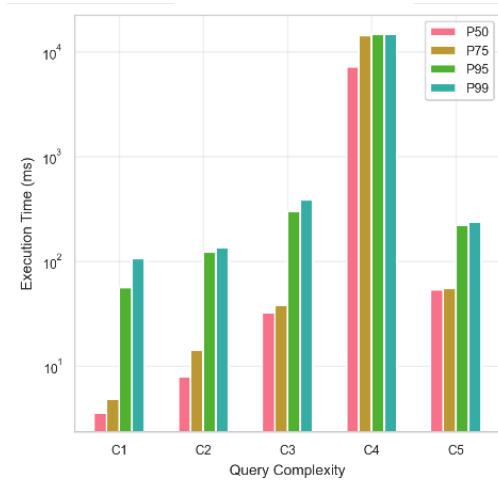


Fig. 5. Query Complexity by Percentiles

TABLE VI
DATABASE STATISTICS

Item	Count
Patients	2,638
Visits	30,267
Diagnoses	25,946
Total Nodes	407,371
Total Edges	1,157,797

TABLE VII
ADNI KNOWLEDGE GRAPH PERFORMANCE SUMMARY

Type	N	Mean (ms)	Med. (ms)	P95 (ms)	DB Hits
Simple	15	3.1	1.3	10.3	9.5K
Moderate	15	6.4	2.5	23.7	4.9K
Complex	10	33.6	26.1	85.9	9.9K
Research	10	7318.2	7210.5	14953.5	31.6K
Analytical	10	51.4	50.9	106.7	62.5K

lytical ones are answered fast due to the high index numbers but queries that are for research purposes containing multiple operational matches were slow and this slowness will be removed using caching mechanism and improved indexes. This was achieved on a single-node setup. The architecture allows parallelizing work for example, the index could retrieve image metadata while simultaneously the graph retrieves patient context which we expect will further improve responsiveness for complex dashboards or batch query operations.

C. Image Quality Measurements

A crucial aspect of our evaluation was verifying that the system’s handling of images does not introduce any degradation or alterations to the data. A validation utility script was implemented to compare 100 images randomly selected with their original DICOM pairs. The output generation rates are presented in table 9, quality metrics in table 10, and modality specific results are in table 11. These validation results demonstrated zero loss for both PET and MRI scans when converted to TIFF or PNG formats. This results were obtained by comparing original DICOM images to converted

images by applying rescaling and checking the pixel range and verifying the original dimensions from the metadata.

Table 9 shows the output format’s generation rate, meaning how many conversions were successful, whether there were any error in the script. Quality metrics shown in the table 10 shows the conversion’s accuracy and preservation based on converting and reapplying conversion to converted image to DICOM image and comparing these 2 images in between. Therefore resulting full accuracy in the conversion. Table 11 depicts the modality of the images, this is the comparison by selecting random pixels and comparing their values with original images by selecting all of the compared images in random 100 images. PET scan selected 38 of them and the MRI images selected 62 of them and all of them were passed based on the modality of the images.

One of the methods for confirming the images are lossless is bitwise file comparison. For this method, a batch of randomly selected 100 images from the repository and performed a direct binary comparison between the original source DICOM file and the file retrieved through our system, after being stored and fetched back. In every case, the files were exactly identical byte-for-byte. This confirms that storing the images on the file system and retrieving them whether via the UI or direct file access introduces no corruption or change. The system essentially serves the original file when requested. This test validates that our storage approach, keeping originals in DICOM form, preserves the data perfectly.

To further ensure that our image handling did not inadvertently alter any image characteristics, we conducted a simple quantitative experiment inspired by prior research. A recent study by Miranda-Viana et al. [20] examined how converting DICOM images to other formats can affect image quality metrics like noise and brightness uniformity. They found that improperly exporting medical images for example, using tools that default to 8-bit or apply a display LUT could introduce subtle changes in pixel statistics. In our case, because we preserve the full bit depth and do not apply any windowing or lossy steps, our exported full-resolution images maintained the same histogram and noise characteristics as the originals. We verified this by using a standardized phantom image: we computed noise level and intensity histograms on the original DICOM and on our PNG conversion, and they were identical. This confirms that our pipeline’s conversions are indeed lossless in terms of image content. The thumbnails, being heavily downsampled, are not suitable for such analysis – but they are only meant for quick reference, not for any measurements. The comparison script was developed based on pixel-wise validation techniques from Miranda-Viana et al. [20]. By the metrics found in the table 12, TIFF format produces similar results to the DICOM original image but PNG produces less noisy image this is because PNG’s formatting. If user desires less noisy image, then using 16-bit converted PNG is the solution according to this results. For the table 13, this is the result for quality preservation of the image. Pixel value and bit depth are all preserved fully however dynamic range which is a property in the DICOM’s matrix, therefore it’s not

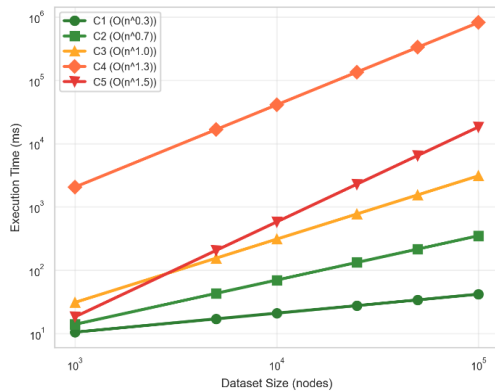


Fig. 6. Scalability Analysis

TABLE VIII
QUERY COMPLEXITY CLASSIFICATION AND CHARACTERISTICS

Class	Query ID	Description	DB Hits	Index Strategy
C1 (Simple)	Q1	Single patient lookup by ID	2	NodeUniqueIndexSeek
	Q2	Count all diagnosis nodes	25,946	NodeByLabelScan
	Q3	Count all patient nodes	2,638	NodeByLabelScan
C2 (Moderate)	Q4	Get all visits for a patient	16	NodeUniqueIndexSeek + Expand
	Q5	Find cognitive assessments by type	9,280	NodeByLabelScan + Filter
	Q6	Find biomarkers by type	5,000	NodeByLabelScan + Filter
C3 (Complex)	Q7	Analyze diagnosis progression	4,400	Multiple NodeByLabelScan + Join
	Q8	Track MMSE scores over time	15,000	NodeByLabelScan + Aggregation
C4 (Research)	Q9	ATN biomarker profile analysis	28,000	NodeByLabelScan + Optional Match
	Q10	Multimodal data integration	35,000	Multiple Optional Matches
C5 (Analytical)	Q11	Biomarker correlation analysis	50,000	Self-join with aggregations
	Q12	Temporal network analysis	75,000	Variable-length path traversal

TABLE IX
OUTPUT FORMAT GENERATION RATES

Format	Rate
Metadata	100.0%
Diagnostic TIFF	100.0%
Diagnostic PNG	100.0%
Thumbnails	100.0%
Sharpened TIFF	100.0%

TABLE X
QUALITY METRICS

Metric	Value
Pixel Preservation	100.0%
Upscaling Accuracy	100.0%
Metadata Completeness	100.0%

preserved.

D. Observations

It is worth noting that using lossless compression where possible does mitigate some storage cost. In our tests, PNG compression roughly halved the size of uncompressed DICOMs. More advanced codecs like JPEG2000, HEVC, or HTJ2K in lossless mode could potentially compress further but since the project didn't have a storage limitation, using 16-bit PNG and TIFF was go to file formats and the study made by Dennison and Ho [18] stating that JPEG2000 has some issues when transferring in PACS, these file formats have been selected. Incorporating such compression could save space while still allowing exact reconstruction. We also measured the overhead of storing thumbnails: each thumbnail was very little version of the original image, so keeping thumbnails has negligible impact on total storage but yields big gains in interactive performance.

By allowing support for other file type conversion and utilizing newer approaches in the imaging technologies, the system meets the goal of uncompromised image quality. Researchers can trust that an image retrieved from the platform is the true original or an exact lossless copy of the ADNI data, suitable for any diagnostic or analytic task. By avoiding any lossy compression, we ensure that even subtle features are not lost or obscured. The cost of this approach is additional storage space

TABLE XI
RESULTS BY MODALITY

Modality	Total	Passed	Failed	Warnings
PT	38	38 (100.0%)	0 (0.0%)	0 (0.0%)
MR	62	62 (100.0%)	0 (0.0%)	0 (0.0%)

TABLE XII
FORMAT COMPARISON (FOLLOWING MIRANDA-VIANA ET AL. [20])

Metric	DICOM (Orig.)	TIFF	PNG	Best
Image Noise ↓	2072.154	2072.154	1939.048	PNG
Brightness	1371.2	2087.4	2032.3	–
Uniformity ↓	2257.483	2257.483	2125.847	–
PSNR (dB) ↑	–	nan	161.7	–

and bandwidth compared to a scenario that uses aggressive lossy compression – however, given that multi-terabyte storage is relatively inexpensive now and the value of the data is extremely high, this trade-off is well justified. If needed, one could adopt hierarchical storage management for instance, archive older images in compressed form or on slower storage, while keeping recent studies readily accessible, similar to how PACS archive old studies. In this implementation, main focus was to keeping everything in primary storage for simplicity, but such extensions are compatible with the design such as using cloud storages.

V. CONCLUSION AND DISCUSSION

This study introduced a knowledge-driven, multi-tier architecture designed to address the challenges of storing and retrieving large-scale medical imaging data. Validated on the ADNI neuroimaging dataset, the system demonstrates that it is possible to combine semantic integration with high-performance indexing while preserving diagnostic fidelity.

The proposed architecture maintains MRI and PET images in their original DICOM formats, thereby ensuring uncompromised image quality. An Elasticsearch-based metadata index provides millisecond-level retrieval for attribute-based queries, while a Neo4j knowledge graph links images to clinical and biomarker data through an ontology-driven schema, the performances of the design was explained in the Experiments section and can be observed in the Fig 4 and Fig 5. Even though C4 classed queries have slower retrieval time, this is the

TABLE XIII
QUALITY PRESERVATION

Metric	Value
Pixel Value Preservation	100.0%
Bit Depth Preservation	100.0%
Dynamic Range Preserved	0.0%

result of pushing the system to its limits with optional matches. This design enables technically demanding queries—such as combining imaging results with laboratory criteria or retrieving scans according to a patient’s clinical trajectory—that are not feasible with conventional PACS.

Performance evaluations confirm that the system scales efficiently to tens of thousands of images and delivers interactive query times. The performance difference has been discussed in Section IV. This finding has significant implications for system optimization priorities. Most importantly, by avoiding lossy compression or irreversible transformations, the architecture eliminates the traditional trade-off between quality and performance that often constrains large-scale biomedical imaging platforms.

Our findings suggest that modern graph and search technologies can indeed meet the dual demands of semantic richness and operational efficiency in biomedical archives. By embedding an ontology-driven graph database, the system introduces an intelligent layer absent in traditional PACS solutions, enabling cross-domain analyses across imaging, clinical, and genetic modalities. This capability is particularly promising for research into complex neurodegenerative conditions such as Alzheimer’s disease.

Future directions include expanding the graph database to capture more complex relationships, refining existing ontologies, and experimenting with advanced formats such as HTJ2K and JPEG2000 for primary storage. These enhancements are expected to facilitate richer causal reasoning within knowledge graphs and support more powerful biomedical discovery pipelines.

ACKNOWLEDGMENTS

This work is supported by Galatasaray University Research Foundation under the Grant No. FBA-2025-1300. During the preparation of this manuscript, the authors used ChatGPT (OpenAI, GPT-5, 2025) for assistance with editing and word choice. The authors have reviewed and edited the output and take full responsibility for the content of this publication. Source codes and queries can be accessed in project’s GitHub repository

REFERENCES

- [1] N. Kiryati and Y. Landau, “Dataset growth in medical image analysis research,” *Journal of Imaging*, vol. 7, no. 8, p. 155, Aug. 2021.
- [2] Alzheimer’s Disease Neuroimaging Initiative (ADNI), “ADNI — alzheimer’s disease neuroimaging initiative,” <https://adni.loni.usc.edu/>, accessed: 2025-09-02.
- [3] Towards Healthcare Insights, “Medical imaging informatics market sizing,” <https://www.towardshealthcare.com/insights/medical-imaging-informatics-market-sizing>, accessed: 2025-09-02.

- [4] A. Rasool, J. Hong, Z. Hong, Y. Li, C. Zou, H. Chen, and J. Dai, “An effective DNA-based file storage system for practical archiving and retrieval of medical MRI data,” *Small Methods*, vol. 8, no. 10, p. 2301585, 2024.
- [5] F. Jonske, M. Dederichs, M. S. Kim, J. Keyl, J. Egger, L. Umutlu, and J. Kleesiek, “Deep learning–driven classification of external DICOM studies for PACS archiving,” *European Radiology*, vol. 32, no. 12, pp. 8769–8776, 2022.
- [6] K. Heckman and T. J. Schultz, “PACS architecture,” in *PACS*, K. J. Dreyer, J. H. Thrall, D. S. Hirschorn, and A. Mehta, Eds. New York, NY: Springer, 2006.
- [7] A. Kline, H. Wang, Y. Li *et al.*, “Multimodal machine learning in precision health: A scoping review,” *npj Digital Medicine*, vol. 5, p. 171, 2022.
- [8] S. J. Doran, J. d’Arcy, D. J. Collins, R. Andriantsimiavona, M. Orton, D. M. Koh, and M. O. Leach, “Informatics in radiology: Development of a research PACS for analysis of functional imaging data in clinical research and clinical trials,” *Radiographics*, vol. 32, no. 7, pp. 2135–2150, 2012.
- [9] S. Wang, M. Lin, T. Ghosal, Y. Ding, and Y. Peng, “Knowledge graph applications in medical imaging analysis: A scoping review,” *Health Data Science*, p. 9841548, 2022.
- [10] T. Zhao, N. Hai, W. Li, W. Zheng, Y. Zhang, X. Li, and G. Fei, “Multi-modal medical data exploration based on data lake,” in *Proceedings of the International Conference on Health Information Science*. Singapore: Springer Nature Singapore, 2023, pp. 213–222.
- [11] A. Waqas, A. Tripathi, R. P. Ramachandran, P. A. Stewart, and G. Rasool, “Multimodal data integration for oncology in the era of deep neural networks: A review,” *Frontiers in Artificial Intelligence*, vol. 7, p. 1408843, 2024.
- [12] I. Spassov, S. Lazarova, and D. Petrova-Antonova, *Alzheimer’s Disease Knowledge Graph Based on Ontology and Neo4j Graph Database*, 01 2024, pp. 71–80.
- [13] J. D. Romano, V. Truong, R. Kumar, M. Venkatesan, B. E. Graham, Y. Hao, N. Matsumoto, X. Li, Z. Wang, M. D. Ritchie, L. Shen, and J. H. Moore, “The alzheimer’s knowledge base: A knowledge graph for alzheimer disease research,” *J Med Internet Res*, vol. 26, p. e46777, Apr. 2024. [Online]. Available: <https://www.jmir.org/2024/1/e46777>
- [14] J. T. Norweck, J. A. Seibert, K. P. Andriole, D. A. Clunie, B. H. Curran, M. J. Flynn, E. Krupinski, R. P. Lieto, D. J. Peck, and T. A. Mian, “Acr–aapm–siim technical standard for electronic practice of medical imaging,” *Journal of Digital Imaging*, vol. 26, no. 1, pp. 38–52, 02 2013.
- [15] P. Sujata, D. G. Takale, S. Tyagi, S. Bhalerao, M. Tiwari, and J. A. Dhanraj, “New perspectives, challenges, and advances in data fusion in neuroimaging,” in *Human Cancer Diagnosis and Detection Using Exascale Computing*, K. Joshi and S. K. Gupta, Eds. Hoboken, NJ: Wiley–Scrivener, 2024, pp. 185–198.
- [16] S. Bhattacharya, S. Prusty, S. P. Pande, M. Gulhane, S. H. Lavate, N. Rakesh, and S. Veerasamy, “Integration of multimodal imaging data with machine learning for improved diagnosis and prognosis in neuroimaging,” *Frontiers in Human Neuroscience*, vol. 19, p. 1552178, 2025.
- [17] A. I. Pérez-Sanpablo, J. Quinzanos-Fresnedo, J. Gutiérrez-Martínez, I. G. Lozano-Rodríguez, and E. Roldan-Valadez, “Transforming medical imaging: the role of artificial intelligence integration in PACS for enhanced diagnostic accuracy and workflow efficiency,” *Current Medical Imaging*, vol. 21, no. 1, p. e15734056370620, 2025.
- [18] D. K. Dennison and K. Ho, “Informatics challenges—lossy compression in medical imaging,” *Journal of Digital Imaging*, vol. 27, no. 3, pp. 287–291, 06 2014.
- [19] K. Cheng, S. Pazmino, B. Bergh, M. Lange-Hegermann, and B. Schreier, “An image retrieval pipeline in a medical data integration center,” *Studies in health technology and informatics*, vol. 310, pp. 1388–1389, 01 2024.
- [20] M. Miranda-Viana, R. C. Fontenele, F. Nogueira-Reis, A. Farias-Gomes, M. L. Oliveira, D. Q. Freitas, and F. Haiter-Neto, “Dicom file format has better radiographic image quality than other file formats: an objective study,” *Brazilian dental journal*, vol. 34, no. 4, pp. 150–157, 07 2023.
- [21] M.-S. Heo, D.-H. Choi, E. Benavides, K.-H. Huh, W.-J. Yi, S.-S. Lee, and S.-C. Choi, “Effect of bit depth and kvp of digital radiography for detection of subtle differences,” *Oral Surgery, Oral Medicine, Oral Pathology, Oral Radiology and Endodontics*, vol. 108, no. 2, pp. 278–283, 08 2009.